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QUATERNIONS AND THEIR GENERALIZATIONS

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1. The discovery of quaternions by W. R. Hamilton in 1843 has led to an extensive theory of linear algebras (or closed systems of hypercomplex numbers) in which the quaternion algebra plays an important rôle. Frobenius[1] proved that the only real linear associative algebras in which a product is zero only when one factor is zero are the real number system, the ordinary complex number system, and the algebra of real quaternions. A much simpler proof has been given by the writer.[2] Later, the writer[3] was led to quaternions very naturally by means of the four-parameter continuous group which leaves unaltered each line of a set of rulings on the quadric surface $x_1^2 + x_2^2 + x_3^2 + x_4^2 = 0$.

The object of the present note is to derive the algebra of quaternions and its direct generalizations without assuming the associative or commutative law. I shall obtain this interesting result by two distinct methods.

2. The term *field* will be employed here to designate any set of ordinary complex numbers which is closed under addition, subtraction, multiplication, and division. Thus all complex numbers form a field, likewise all real numbers, or all rational numbers.

Just as a couple (a, b) of real numbers defines an ordinary complex number $a + bi$, where $i^2 = -1$, so also an n-tuple $(x_1, \ldots, x_n)$ of numbers of a field F defines a hypercomplex number

$$x = x_1 e_1 + x_2 e_2 + \ldots + x_n e_n, \tag{1}$$

where the *units* $e_1, \ldots, e_n$ are linearly independent with respect to the field F and possess a multiplaction table

$$e_i e_j = \sum_{k=1}^{n} \gamma_{ijk} e_k \qquad (i, j = 1, \ldots, n), \tag{2}$$

n which the γ's are numbers of F. Let $x' = \Sigma x_i' e_i$ be another hypercomplex number whose coördinates x_i' are numbers of F. Then shall

$$xx' = \sum_{i,j=1}^{n} x_i x_j' . e_i e_j, \quad x \pm x' = \sum_{i=1}^{n} (x_i \pm x_i') e_i, \quad fx = xf = \sum_{i=1}^{n} f x_i . e_i,$$

when f is in F, so that multiplication is distributive. Under these assumptions, the set of all numbers (1) with coördinates in F shall be called a *linear algebra over* F.

3. We assume that e_1 is a principal unit (modulus), so that $e_1 x = x e_1 = x$ for every number x of the algebra, and write 1 for e_1. We assume that every number of the algebra satisfies a quadratic equation with coefficients in F. If $e^2 + 2ae + b = 0$, $(e + a)^2 = a^2 - b$, so that we may take the units to be $1, E_2, \ldots, E_n$, where $E_i^2 = s_{ii}$, a number of F. For i and j distinct and >1, $E_i \pm E_j$ satisfies a quadratic, so that $(E_i \pm E_j)^2 = s_{ii} + s_{jj} \pm (E_i E_j + E_j E_i)$ is a linear function of $E_i \pm E_j$. Thus $E_i E_j + E_j E_i$ is a linear function of $E_i + E_j$ and of $E_i - E_j$, and hence is a number $2s_{ij} = 2s_{ji}$ of F.

Let $u_2, \ldots, u_n$ be arbitrary numbers of F and write $U = \Sigma u_k E_k$. Then U^2 equals

$$Q = \sum_{k,l=2}^{n} s_{kl} u_k u_l .$$

It is a standard theorem that Q can be reduced to $\Sigma c_i v_i^2$ by a linear transformation $u_k = \Sigma a_{kl} v_l$ with coefficients in F and of determinant $\neq 0$. Write

$$e_l = \sum_{k,l=2}^{n} a_{kl} E_k \qquad (l = 2, \ldots, n).$$

Then $1, e_2, \ldots, e_n$ are linearly independent and may be taken as the new units of our algebra over F. Then

$$U = \sum_{k,l} a_{kl} v_l E_k = \sum_{l} v_l e_l,$$

$$U^2 = \sum_{k,l} v_k v_l e_k e_l = \sum c_i v_i^2.$$

Hence

$$e_i^2 = c_i, \; e_i e_j + e_j e_i = 0 \qquad (i, j = 2, \ldots, n, \; i \neq j). \tag{3}$$

4. Write $x = x_1 + \Sigma x_i e_i$. Then $(x - x_1)^2 = \Sigma c_i x_i^2$. This gives $xx' = x'x = \sigma$, where

$$x' = 2x_1 - x = x_1 - \sum_{i=2}^{n} x_i e_i, \quad \sigma = x_1^2 - \sum_{i=2}^{n} c_i x_i^2.$$

We shall call x' the *conjugate* to x and $\sigma = \sigma(x)$ the *norm* of x. Hence the product of any number and its conjugate in either order equals its norm. We assume that the norm of a product equals the product of the norms of the factors:

$$\sigma(x)\sigma(\xi) = \sigma(X), \text{ if } x\xi = X, \tag{4}$$

and shall investigate the resulting types of linear algebras. We assume also that each $c_i \neq 0$ in (3).

5. By (2) the coördinates of $X = x\xi$ are $X_k = \Sigma x_i \xi_j \gamma_{ijk}$. Since $e_i^2 = c_i$, we have $\gamma_{i1} = c_i$, $\gamma_{ik} = 0$ $(i>1, k>1)$. Hence

$$X_1 = x_1\xi_1 + \sum_{i=2}^{n} x_i\xi_i c_i + \sum_{\substack{i,j=2 \\ i \neq j}}^{n} x_i\xi_j\gamma_{ij1},$$

$$X_k = x_1\xi_k + x_k\xi_1 + \sum_{\substack{i,j=2 \\ i \neq j}}^{n} x_i\xi_j\gamma_{ijk} \quad (k>1) \tag{5}$$

Since this transformation is the identity $X = x$ if $\xi = 1$, we obtain an infinitesimal transformation by taking $\xi_1 = 1$, $\xi_j = \delta t$, $\xi_i = 0 (i \neq 1, j)$:

$$\delta x_1 = X_1 - x_1 = \left\{ c_j x_j + \sum_{\substack{i=2 \\ i \neq j}}^{n} \gamma_{ij1} x_i \right\} \delta t, \delta x_j = \left\{ x_1 + \sum_{\substack{i=2 \\ i \neq j}}^{n} \gamma_{ijj} x_i \right\} \delta t,$$

$$\delta x_k = \left\{ \sum_{\substack{i=2 \\ i \neq j}}^{n} \gamma_{ijk} x_i \right\} \delta t \qquad (k \neq 1, j). \tag{6}$$

For these ξ's, $\sigma(\xi)$ is unity to within an infinitesimal of the second order. Hence the increment to $\sigma(x)$ must vanish identically, so that

$$\gamma_{ij1} = \gamma_{ijj} = \gamma_{iji} = 0 \qquad (1, i, j \text{ distinct}), \tag{7}$$

$$c_k\gamma_{ijk} + c_i\gamma_{kji} = 0 \qquad (1, i, j, k \text{ distinct}). \tag{8}$$

By (7), (5) simplifies to

$$X_1 = x_1\xi_1 + \sum_{i=2}^{n} x_i\xi_i c_i, \; X_k = x_1\xi_k + x_k\xi_1 + \sum x_i\xi_j\gamma_{ijk} \; (k>1), \tag{9}$$

where, in the final sum, i and j range over distinct values from 2, ..., n, excluding k. This final sum is, therefore, absent if[4] $n=3$; whence $\sigma(X)$

has the term $2x_2\xi_2c_2.x_3\xi_3c_3$ which does not occur in $\sigma(x)\sigma(\xi)$. But $c_2c_3 \neq 0$ by hypothesis. Hence $n>3$.

Hitherto we have not examined the conditions which follow from the final equations (3); these are

$$\gamma_{jjk} = -\gamma_{ijk} \quad (i, j = 2, \ldots, n; i \neq j). \tag{10}$$

6. Taking $n = 4$ and applying (10), we see that (9) become

$$\begin{cases} X_1=x_1\xi_1 + c_2x_2\xi_2 + c_3x_3\xi_3 + c_4x_4\xi_4, X_2=x_1\xi_2 + x_2\xi_1 + \gamma_{342}(x_3\xi_4-x_4\xi_3), \\ X_3=x_1\xi_3 + x_3\xi_1 + \gamma_{243}(x_2\xi_4-x_4\xi_1), X_4=x_1\xi_4 + x_4\xi_1 + \gamma_{234}(x_2\xi_3-x_3\xi_2). \end{cases} \tag{11}$$

These transformations do not in general form a group and hence are not generated by the corresponding infinitesimal transformations employed above. Hence it remains to require that $\sigma(X) = \sigma(x)\sigma(\xi)$ under the transformations (11). The conditions are seen to be

$$c_3c_4 = -c_2\gamma^2_{342}, c_2c_4 = -c_3\gamma^2_{243}, c_2c_3 = -c_4\gamma^2_{234}, c_4\gamma_{234} = c_2\gamma_{342} = -c_3\gamma_{243},$$

the first two of which reduce to the third by means of the last three equations. To these last can be reduced all the conditions (8) by means of (10).

Applying the transformation of variables which multiplies x_4, ξ_4, X_4 by γ_{234}, and leaves the remaining x_i, ξ_i, X_i unaltered, we get

$$\begin{cases} X_1=x_1\xi_1 + c_2x_2\xi_2 + c_3x_3\xi_3 - c_2c_3x_4\xi_4, X_2=x_1\xi_2 + x_2\xi_1 - c_3x_3\xi_4 + c_3x_4\xi_3, \\ X_3=x_1\xi_3 + x_3\xi_1 + c_2x_2\xi_4 - c_2x_4\xi_2, \quad X_4=x_1\xi_4 + x_4\xi_1 + x_2\xi_3 - x_3\xi_2. \end{cases} \tag{11^1}$$

These are the values obtained by Lagrange[5] in his generalization $\sigma(x)\sigma(\xi) = \sigma(X)$ of Euler's formula for the product of two sums of four squares. Then $x\xi = X$ gives the following multiplication table for the units:

$$\begin{cases} e_2^2 = c_2, e_3^2 = c_3, e_4^2 = -c_2c_3, e_2e_3 = e_4, e_3e_2 = -e_4, \\ e_2e_4 = c_2e_3, e_4e_2 = -c_2e_3, e_3e_4 = -c_3e_2, e_4e_3 = c_3e_2. \end{cases} \tag{12}$$

This algebra is associative and is the direct generalization of quaternions to a general field F which the writer[6] obtained elsewhere from assumptions including associativity. The four-rowed determinants of the general number x of this algebra equals $\sigma^2(x)$. The case $c_2 = c_3 = -1$ gives the algebra of quaternions, for which it is customary to write i, j, k instead of our units e_2, e_3, e_4.

7. It is not very laborious to show by the above method that the cases $n = 5$ and $n = 6$ are excluded. However, Hurwitz has proved that a relation of the form $\sigma(x)\sigma(\xi) = \sigma(X)$ is impossible if $n \neq 1, 2, 4, 8$. A slight simplification of his proof, together with an account of the history of this problem, has been given by the writer.[7] Hurwitz made no attempt to find all solutions when $n = 4$. We proceed to treat this problem.

Consider the case $c_i = -1$ to which the general case may be reduced by an irrational transformation. Then $\sigma(x) = \Sigma x_i^2$. We investigate the linear algebras having property (4), i. e.,

$$(x_1^2 + \ldots + x_n^2)(\xi_1^2 + \ldots + \xi_n^2) = X_1^2 + \ldots + X_n^2, \tag{13}$$

if

$$X_k = \sum_{j=1}^{n} \Big(\sum_{i=1}^{n} \gamma_{ijk}\, x_i \Big) \xi_j \qquad (k = 1, \ldots, n). \tag{14}$$

The matrix M of this substitution has the element $\Sigma_{i=1}^{i=n}\gamma_{ijk}x_i$ in the kth row and jth column. If this substitution is applied to a quadratic form in $X_1, \ldots, X_n$ of matrix Q, it is a standard theorem that we obtain a quadratic form in $\xi_1, \ldots, \xi_n$, whose matrix is $M'QM$, where M' is the transposed of M, being obtained from M by the interchange of its rows and columns. In our problem, Q is the identity matrix I whose elements are all zero except the diagonal elements which are 1. Hence, by (13),

$$M'M = (x_1^2 + \ldots + x_n^2)I. \tag{15}$$

When a homogeneous polynomial $\sigma(x_1, \ldots, x_n)$ of any degree has the property (4) of possessing a theorem of multiplication, the writer[8] has proved that we may apply a linear transformation on $x_1, \ldots, x_n$ which leave $\sigma(x)$ unaltered and one on $\xi_1, \ldots, \xi_n$ which leaves $\sigma(\xi)$ unaltered such that the new algebra has the principal unit e_1, so that γ_{1jk} and γ_{j1k} are both 0 if $j \neq k$, and both unity if $j = k$.

Hence $M = x_1M_1 + \ldots + x_nM_n$, where γ_{ijk} is the element in the kth row and jth column of M_i, whence $M_i = I$. Thus (15) gives

$$M_i' = -M_i,\ M_i'M_i = I,\ M_i'M_j + M_j'M_i = 0 \quad (i>1, j>1, j \neq i). \tag{16}$$

In view of the values of γ_{jik}, and $M_i' = -M_i$, we have, when $n = 4$,

$$M_2 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -\gamma_{223} & -\gamma_{224} \\ 0 & \gamma_{223} & 0 & -\gamma_{234} \\ 0 & \gamma_{224} & \gamma_{234} & 0 \end{pmatrix}, \qquad M_3 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & -\gamma_{323} & -\gamma_{324} \\ 1 & \gamma_{323} & 0 & -\gamma_{334} \\ 0 & \gamma_{324} & \gamma_{334} & 0 \end{pmatrix},$$

$$M_4 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -\gamma_{423} & -\gamma_{424} \\ 0 & \gamma_{423} & 0 & -\gamma_{434} \\ 1 & \gamma_{424} & \gamma_{434} & 0 \end{pmatrix}.$$

By $M_i'M_i = I$, we have $M^{2i} = -I$, which gives

$$\gamma_{223} = \gamma_{224} = \gamma_{323} = \gamma_{334} = \gamma_{424} = \gamma_{434} = 0,\ \gamma^2 = \delta^2 = \epsilon^2 = 1,$$

where $\gamma = \gamma_{234}$, $\delta = \gamma_{324}$, $\epsilon = \gamma_{423}$. Hence

$$M_2 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\gamma \\ 0 & 0 & \gamma & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -\delta \\ 1 & 0 & 0 & 0 \\ 0 & \delta & 0 & 0 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -\epsilon & 0 \\ 0 & \epsilon & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{17}$$

The final condition (16) states that M_iM_j is skew-symmetric. The products M_2M_3, M_2M_4, M_3M_4 of matrices (17) are seen at once to be

skew-symmetric if and only if $\delta = -\gamma$, $\epsilon = \gamma$, and then $M_2M_3 = \gamma M_4$, $M_2M_4 = -\gamma M_3$, $M_3M_4 = \gamma M_2$. Writing i, j, k for $M_2, M_3, \gamma M_4$, we have the multiplication table of quaternions. Or we may form the matrix M and write X_k for the sum of the products of the elements of its kth row by $\xi_1, \ldots, \xi_4$, and take $\gamma = 1$ (by multiplying x_4, ξ_4, X_4 by γ); we obtain (11′) for $c_2 = c_3 = -1$. Hence we have again obtained the quaternion algebra without assuming the associative law. The case $n = 8$ is being investigated in this way by one of my students.

[1] Frobenius, *Jour. für Math.*, **84**, 1878 (59).

[2] Dickson, *Linear Algebras*, "Cambridge Tracts in Mathematics and Mathematical Physics," No. **16**, 1914 (10–12).

[3] Dickson, *Bull. Amer. Math. Soc.*, **22**, 1915 (53–61).

[4] By (7), $e_2e_3 = e_3e_2 = 0$.

[5] Lagrange, *Nouv. Mém. Acad. Roy. Sc. de Berlin*, année 1770, Berlin, 1772 (123–133); *Oeuvres* de Lagrange, **3**, 1869 (189). Reproduced in Dickson's *History of the Theory of Numbers*, II, 1920 (279–281).

[6] Dickson, *Trans. Amer. Math. Soc.*, **13**, 1912 (65).

[7] Dickson, *Annals of Math.*, **20**, 1919 (155–171, 297).

[8] Dickson, *Comptes Rendus du Congrès Internat. Math.*, Strasbourg, 1920 (131–146).

NOVOCAINE AS A SUBSTITUTE FOR CURARE[1]

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Communicated by G. H. Parker, March 3, 1921

Since the recent war, the need of a substitute for the Indian arrow poison, curare, has been keenly felt in many physiological laboratories. While investigating the activity of certain local anesthetics, it was found that novocaine, in its effect upon the neuro-muscular mechanism of frogs, duplicates in many particulars the unique action of curare.

If the sciatic nerve of a sciatic-gastrocnemius preparation is bathed in a strong solution of novocaine (2.5 per cent in water or in physiological salt solution) for as long as twenty minutes, no decrease in its conductivity can be observed. However, if the muscle itself is bathed in such a solution (by direct immersion or, "painting" with a camel's hair brush) the power of reacting to nervous stimulation is destroyed within three to five minutes, though ability to respond by contraction to direct electrical stimulation remains unimpaired. Thus, in the action of novocaine there is a complete duplication of the properties originally described by Claude Bernard for curare.

Whether novocaine acts directly upon the end-plates of the motor fibers or upon some membrane intermediate between the plates and the

[1] Contributions from the Zoölogical Laboratory of the Museum of Comparative Zoölogy at Harvard College. No. 330.

muscle has not been determined. A dye that I have made by linking novocaine with a benzene nucleus was found to be physiologically active, like the 2.5% novocaine, and appears to stain the elements acted upon by the novocaine. When the stain is diazotized directly into the living muscle, by putting the tissue first into novocaine and then into a solution of the staining base, only the muscle nuclei take the stain deeply, the nervous elements of the end-plates as well as the motor fibers remaining uncolored. It seems reasonably certain, therefore, that novocaine acts upon some constituent of the neuro-muscular mechanism beyond the end-plates. The significance of the affinity of the dye for the muscle nuclei is as yet unknown. The object of this note is to direct the attention of physiologists to a convenient substitute for curare.

NOTE ON A METHOD OF DETERMINING THE DISTRIBUTION OF PORE SIZES IN A POROUS MATERIAL

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Communicated by W. A. Noyes, February 12, 1921

The pressure required to force mercury into a capillary pore of radius, r, is $\frac{-2\gamma\cos\theta}{r}$, where γ is the surface tension and θ the angle of contact. Upon this relation can be based a method for determining the effective pore diameters in a porous material such as charcoal. If pores of various diameters are present, one may determine also the fraction of the total porosity which is due to pores having effective diameters lying between any two stated limits. The procedure would be as follows:

The coarsely granular sample of the thoroughly outgassed material is weighed and placed in a steel pressure bomb which is then evacuated until all adsorbed gases are removed. Pure mercury is then admitted to fill the bomb and a series of pressure and volume measurements are made at various pressures up to the highest pressure it is desired to employ.

The decrease in volume, ΔV, accompanying a small pressure increase of Δp, in any part of the range must evidently be due to the filling of pores whose effective radii lie between the limits r and $r - \Delta r$, or

$$\frac{\Delta r}{\Delta p} = \frac{-2\gamma\cos\theta}{p^2}.$$

A blank experiment without the porous material should of course be made in order to correct for the compressibility of the mercury and the expansion of the bomb under pressure. For accurate results the compressibility of the porous material, and the variation of γ and θ with p should also be known.

The value of θ could be determined from an X-ray photograph of a mercury meniscus in a capillary composed of the material under examination, or by the drop-shape method.

AN APPARENT CASE OF NON-MENDELIAN INHERITANCE IN DATURA DUE TO A DISEASE

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Communicated by C. B. Davenport, February 26, 1921

In the common Jimson Weed (*Datura Stramonium*), spiny or armed capsules are dominant to smooth or *inermis* capsules. A cross between two plants, each homozygous for a single member of this pair of characters will produce spiny fruited offspring in the F_1 and a ratio of 3 spiny to 1 smooth plant in the F_2 generation.

In 1915, the writer found a single *inermis* plant in a culture where smooth capsules could not have occurred through segregation. It was considered a new *inermis* mutation and its inheritance was therefore studied in crosses with normals.

The new form was called Quercina on account of the increased oak-like dentation of its leaves. The most conspicuous character on the mature plant was the partial or complete suppression of spines on the capsules. An examination of the plants throughout the growing and flowering condition indicated that other parts of the plant were also involved and showed such changes as the slitting of the normally undivided corolla, the absence of pollen, which caused the plant to depend upon outcrossing in order to set seed, and certain other characters associated with less vigorous growth.

Later investigation showed that the Quercina character occurred spontaneously in the garden cultures in many ways like a vegetative mutation. In a single year's test, about $1^1/_4$ per cent of the normal plants in the field took on the Quercina character by the last of the season. This Quercina character generally shows itself weakly in a single branch and gradually spreads to all the new growth. It occasionally happens on plants which are acquiring the Quercina character that capsules will be found in a transitional condition with their spines only partially reduced. Sometimes some of the valves may be entirely smooth and others on the same capsule fully spined.

So far as we can judge from the literature, other investigators who have worked on the Jimson Weed found have Quercina plants in their cultures. They seem not to have noticed, however, any of the distinctive characters other than those shown by the capsules.

Naudin[1] reports in an F_1 between an *inermis* variety (*D. laevis*) of *D. Stramonium* and the armed type, that while most of the plants had spiny fruits, others had fruits with reduced spines. Many of the capsules on three out of the forty plants in this generation were very spiny on part of the surface while totally smooth on the rest. Naudin believed that they united thus by distinct and separate compartments the distinctive traits of the parental types *D. laevis* and *D. Stramonium*. He calls this "*hybridité disjointe*" and cites in this connection the condition in the graft chimera *Cytisus Adami*. In a series of F_2 plants from the same original cross Naudin found 6 individuals out of 38 which again showed more or less well marked his "*hybridité disjointe*" and which were presumably Quercinas.

Godron[2] reports finding capsules partly spiny and partly smooth. He objects, however, to Naudin's interpretation that the separation of the fruit into smooth and spiny portions is due to their origin from smooth and spiny parents, since he says that he has found this condition when *both* parents had spines.

Bateson and Saunders,[3] after discussing the intermediate color of the flowers in the F_1 between white and purple flowered forms, say: "The occurrence of intermediate forms was also occasionally noticeable in the fruits. Among the large number of capsules examined, there were some of the mosaic type, in which part of the capsule was prickly and the remainder smooth, while others suggesting a blend, were more or less prickly all over, but the prickles were much reduced in size, and often formed mere tubercles. These mosaics occurred as rareties both on prickly individuals and on smooth ones still more rarely." Further evidence pointing to the Quercina nature of these abnormal capsules is given in their following statement: "Such intermediate fruits were most often found towards the end of the flowering season."

It may be stated that the writer has grown many thousand plants of the Jimson Weed, including large numbers of individuals that were heterozygous for *inermis* capsules, but has never observed any except Quercinas which showed a mosaic arrangement of the spines on the fruits, with some valves smooth and others more or less spiny. Despite their misconception of the nature of the phenomenon which they observed, it is highly probable that the investigators mentioned were describing the same type of plant that we have considered under the term Quercina.

The Quercina character does not breed true. It is transmitted by seed to about 79 per cent of its offspring when pollinated from normal plants. The remaining 21 per cent normal offspring do not produce Quercina seedlings in the next generation, and hence do not carry the character in a recessive condition.

Two plants which were changing over from a normal to a Quercina condition were investigated with regard to the type of offspring obtained from their individual capsules. There seems to be only a rough correspondence between the strength of the Quercina character in the parent and the number of Quercina plants in its offspring.

As has been stated already, Quercina flowers are characteristically devoid of pollen. In a few plants, however, which were becoming transformed into Quercinas, pollen was obtained and used in crosses onto normal plants. Quercina offspring were obtained from such crosses showing that the character in question can be transmitted through the male.

The non-Mendelian nature of the breeding results with Quercina plants as well as the spontaneous occurrence of the character on branches of normal plants and its spread to the new growth suggested the desirability of attempting to transmit the character by grafting. This attempt was entirely successful and Quercina cions grafted onto normal plants of the Jimson Weed causes the new growth of the stock to take on the appearance of Quercina. There is ample evidence, therefore, to indicate that the cause of the Quercina complex, which by ourselves as well as by earlier investigators had been supposed to be a manifestation of a genetic character, is in fact a disease transmissible by grafting.

It has not been found possible to infect plants artificially by rubbing with diseased leaves nor by inoculation of expressed juice from Quercina plants.

The profound morphological changes brought about in the leaves and especially in the flowers and fruit are such that Quercina individuals would be considered worthy of specific if not of generic separation if 100 per cent of the seedlings instead of only 79 per cent came true to the Quercina complex. As the facts stand, however, there is much in the behavior of Quercina plants which suggests genetic phenomena.

To one who has read the data presented in the foregoing pages, certain similarities may be apparent between the Quercina in the Jimson Weed and rogues in peas investigated by Bateson and Pellew.[4] It will not be desirable, however, to discuss the two forms more in detail until grafting experiments with the rogues in peas have given definite results.

A detailed account of the Quercina disease with photographs will appear shortly in the *Journal of Genetics*.

[1] Naudin, Ch., Nouvelles Recherches sur L'Hybridité dans les Végétaux, *Nouv. Arch. Mus.*, **1**, 1865 (41–54).

[2] Godron, D. A., *Des hybrides et des metis de Datura.*, Nancy, 1873 (1–75).

[3] Bateson, W. and Saunders, E. R., Report to Evolution Committee of the Royal Society, **1**, 1902 (21–32).

[4] Bateson, W. and Pellew, Caroline, "The Genetics of 'Rogues' among Culinary Peas," *Proc. Royal Soc.*, **91**, 1920 (186–195).

THE EFFECT OF A VARIATION IN MILLING ON THE DIGESTIBILITY OF WHEAT FLOURS

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CONTRIBUTION FROM THE OFFICE OF HOME ECONOMICS, U. S. DEPARTMENT OF AGRICULTURE

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Introduction.—Cereals are almost universally the great source of carbohydrate food in the human ration, and with moderate additions of fat and protein they are capable of sustaining life and health over long periods of time. They may be stored for years with little deterioration in quality, are easily transported and yield the greatest amount of food for the labor expended in producing them. Of the cereal grains, wheat, largely because of its gluten content which causes it to produce superior bread making flour, has come to be considered as an almost indispensable article of diet throughout Western Europe and America. The normal, annual per capita consumption of wheat in this country is approximately five bushels.[1] Valuable as wheat is in the human dietary in normal times, it is even more important in times of stress, for a disturbance in the bread supply has an immediate and often harmful effect on national morale.

The shortage in the food supply of the world in 1917 made the collection and correlation of data concerning human nutrition of utmost importance and many matters were re-examined in the hope of providing additional data of value. Extensive investigations of the chemical composition, digestibility, and nutritive value of wheat and wheat products have been made by European investigators and in the United States. Earlier investigations of this Office, undertaken with the coöperation of the Maine and Minnesota Agricultural Experiment Stations, included studies of the digestibility of wheat flours prepared under a variety of milling conditions from wheat grown under uniform climatic, seasonal and soil conditions. The results[2] obtained showed that the protein of white flour (72% extraction) was 88.1% digested, that of entire wheat flour (85% extraction) was 81.9% digested, and that of graham flour (100% extraction) was 76.9% digested. While the data were sufficient to permit general conclusions regarding the effect of variation of milling on digestibility, the unprecedented world shortage of food during the World War made rigid economy in the use of food materials necessary and in 1917 the U. S. Food Administration requested this Office to secure additional data concerning the relative digestibility of flours representing a series of extractions from the same lot of wheat. The results obtained from the work thus undertaken were available to the Food Administration during the war period. The findings, however, are not limited in their application to special conditions under which the work was carried out but

are of equal interest in the consideration of general problems of nutrition and the economic utilization of the food supply under normal conditions.

The flours studied were prepared from the wheat mixture furnished to the flour millers by the U. S. Grain Corporation late in 1917. It consisted of the following:

20% choice hard spring (largely Marquis)
25% 58 lbs. spring
15% velvet chaff
25% slightly smutty spring
10% durum
5% Kansas and Oklahoma

At the request of the Food Administration the wheat was so milled by commercial flour mills that a uniform series of extractions was obtained. As required by the U. S. Food Administration regulations then in force, the amount of wheat used was limited in the proportion of 264 lbs. of 58-lb. wheat for 196 lbs. of flour. The milling data follow:

75% out of	72% extraction yielding	54% of the wheat
95% out of	74% extraction yielding	70% of the wheat
100% out of	85% extraction yielding	85% of the wheat
100% out of	100% extraction yielding	100% of the wheat

For the sake of convenience the above flours will be referred to according to the percentage of the wheat extracted. Described in ordinary commercial terms, the 54% flour contained only patent flour. The 70% extraction, that is 95% of 74% extraction, is what is known commercially as a 95% patent, or "standard patent" and contained patent, 1st clear, and a small portion of the 2nd clear. The 85% flour corresponds to the old-fashioned "whole-wheat" flour, and contained patent, 1st clear, 2nd clear, Red Dog, and Shorts. The 100% flour corresponds to that known formerly as graham flour and contained the entire wheat kernel.

These experiments were begun before the Food Administration had adopted the milling of what was later known as "war flour." The latter was sometimes known as "100% war flour," but this did not mean that it contained 100% of the wheat. In milling the "war flour," 100 lbs. of cleaned wheat was separated into two products: about 75 lbs. of flour (never less than 74.3 lbs.) and about 25 lbs. of feed, giving a flour of about 75% extraction. The 95% patent or 70% extraction flour used in these experiments was similar to the "war flour" save that 5% of low grade and 2nd clear stock had been removed to improve its bread-making quality.

Feeding experiments were conducted with healthy young men according to the method commonly used by the Office of Home Economics in studies of digestibility.[3]

In order to ascertain whether the subjects acquired during the period of the experiments any increased tolerance for the coarser flours the squad

was divided into groups of five subjects each, and one group was served the 54%, 70%, 85%, and 100% flours in the order given while the second group was served the flours in the reverse order. The results of the 139 digestion experiments reported below do not indicate that the first group of subjects acquired any tolerance for the coarser flours not possessed by the second group which was served the coarser flours first.

The experimental diet was so chosen that its preparation should involve a minimum of labor, that the bread should supply the larger portion of the total protein, and that it should be representative of a simple mixed diet. The diet consisted of bread, fruit (orange), butter, sugar, with tea or coffee as a beverage if desired.

The fruit, butter, and sugar were served as purchased from a nearby market. A quantity of bread sufficient to supply all the subjects for one day was prepared daily, the ingredients being used in the following proportion:

3 cups sifted flour
1 teaspoon salt
1 tablespoon sugar
1/8 to 1/2 cake yeast
1 tablespoon hydrogenated vegetable oil
1 cup water

While data relative to the digestibility of the diet as a whole are of value, interest was primarily centered in the digestibility of the bread or rather the flour from which the bread was made. The digestibility of the protein and carbohydrate supplied by the different breads has been estimated by making a correction for the undigested protein and carbohydrate resulting from the accessory foods by a method outlined in detail in earlier publications.[4] It has been assumed from the results of the early investigations of this Office[5] that the digestibility of the protein of butter is 97% and protein of fruit, 85%; of the carbohydrate of fruit, 90%; and of sugar, 98%.

In planning the investigation, it was decided to make a sufficient number of digestion experiments so that the average results obtained should be of general application. The experimental periods were to be long enough so that the effect of any irregularity in the rate of passage of food residues through the alimentary tract, and any error in the separation or collection of feces would be practically negligible. The experimental periods which were from 15 to 25 days in length were subdivided and considered as separate three-day experiments, following one directly after the other.

The subjects for the experiments were students in local educational institutions, and were selected with care in order that the conclusions might be applicable to the average normal adult. An attempt was made to include in the squad some accustomed to strenuous exercise, some accustomed to take light exercise, and some who took little recreation, in

order to have under observation individuals whose peristaltic action was normal and regular and others who had a tendency toward constipation.

Summary of Results.—The results obtained in the studies of 54%, 70%, 85%, and 100% flours are summarized in the table which follows:

TABLE 5

SUMMARY OF EXPERIMENTS ON THE DIGESTIBILITY OF WHEAT FLOURS

KIND OF FLOUR USED IN PREPARATION OF BREAD	NUMBER OF EXPTS.	AVERAGE AMOUNT OF BREAD EATEN PER MAN PER DAY	DIGESTIBILITY OF ENTIRE RATION			DIGESTIBILITY OF BREAD PROTEIN	DIGESTIBILITY OF BREAD CARBOHYDRATE
			PROTEIN	FAT	CARBOHYDRATE		
		GRAMS	PER CENT	PER CENT	PER CENT	PER CENT	PER CENT
54% flour	43	600	87.8	96.5	98.8	87.7	99.7
70% flour	42	564	90.1	96.1	99.0	90.1	99.9
85% flour	21	472	87.1	96.9	97.5	87.1	98.5
100% flour	33	663	84.2	93.9	95.0	84.2	94.4

In the above experiments the subjects ate on an average considerably more than a pound of bread daily for periods of from 6 to 20 days without producing any digestive disturbances, which indicates that wheat flours, regardless of percentage of extraction, are well tolerated by the human body.

The coefficients of digestibility, obtained from the 139 digestion experiments reported above, 87.7% for protein and 99.7% for carbohydrate of 54% flour; 90.1% for protein and 99.9% for carbohydrate of 70% flour (95% patent); 87.1% for protein and 98.5% for carbohydrate of 85% ("whole wheat") flour; and 84.2% for protein and 94.4% for carbohydrate of 100% flour, show these flours to be well digested. From these results it appears that the U. S. Food Administration in attempting to obtain efficient utilization of the wheat supply secured did well to specify that wheat should be milled at 75% extraction, that is an extraction similar to the one which in these experiments showed the highest proportion of digestible nutrients.

The digestibility of the fat content of the experimental diet was quite uniform and was practically identical with that of butter and "shortening"[6] which comprised the major portion of the fat consumed, except in the series of experiments with 100% flour in which the fat was only 93.7% digested.

Attention was given to the effect of the different flours on peristaltic action. The 54% and 70% flours did not tend to produce constipation during a period of 15 to 18 days, and although a somewhat freer movement of the bowels resulted from the continued use of 85% and 100% flours no case of pronounced laxative effect was noted.

In general the results of the digestion experiments here reported are in accord with conclusions drawn from earlier studies of the digestibility of wheat flours. The digestibility of the 70% (95% patent) flour was the highest, that of the 54% flour was slightly greater than that of the 85% ("whole wheat") flour, while the digestibility of the 100% (graham) flour was lowest of all those studied. Since the flavor of bread made with the different flours varies, the use of different kinds for bread making is an easy way of giving variety to the diet.

[1] U. S. Dept. Agr., Bur. Crop Estimates, *Monthly Crop Rept.*, **3**, 1917, No. 10 (99).

[2] U. S. Dept. Agr., *Office Expt. Sta. Bull.* **85**, 1900 (32–33); *Bull.* **101**, 1901 (33); *Bull.* **126**, 1903 (29, 45); *Bull.* **143**, 1904 (32); *Bull.* **156**, 1905 (36).

[3] U. S. Dept. Agr., *Bull.* **310**, 1915; **617**, 1919; **717**, 1919.

[4] U. S. Dept. Agr., *Bull.* **470**, 1916 (7); **525**, 1917 (4).

[5] *Connecticut Storrs Sta. Rpt.*, 1899 (104).

[6] U. S. Dept. Agr., *Bull.* **310**, 1915.

THE MATHEMATICAL THEORY OF THE APPORTIONMENT OF REPRESENTATIVES[1]

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The Problem.—The exact quota to which each state is theoretically entitled on the basis of population usually involves a fraction. *The problem is, to replace these exact quotas by whole numbers* in such a way that the resulting injustice (due to adjustment of the fractions) shall be as small as possible.

This problem has been the subject of violent debate in Congress for the past one hundred years, a new method of apportionment having been proposed after almost every decennial census. None of these methods, however, possesses any satisfactory mathematical justification. The need of a strictly mathematical treatment of the problem having been called to the writer's attention by Dr. J. A. Hill, Chief Statistician of the Bureau of the Census, the following solution has been worked out on the basis of two very simple postulates. The new method may be called the *Method of Equal Proportions*.

Let N be the total number of representatives, $A, B, C, \ldots$ the populations of the several states, and $a, b, c, \ldots$ the number of representatives assigned to each.

Fundamental Principle.—In a satisfactory apportionment between two states (A greater than B), we shall agree that A/a and B/b should be as nearly equal as possible; also a/A and b/B; also A/B and a/b; also B/A and b/a.

Now to say that two quantities are "nearly equal" may be interpreted to mean: *either*, that the difference between the quantities is nearly zero; *or*, that the ratio between them is nearly one.

(Here the difference "between" two quantities means the larger minus the smaller. Similarly, the ratio "between" two quantities means the larger divided by the smaller.)

If we adopt the "difference" interpretation, we have:

POSTULATE Ia. *The difference between A/a and B/b;* or

POSTULATE Ib. *The difference between a/A and b/B;* or

POSTULATE Ic. *The difference between A/B and a/b;* or

POSTULATE Id. *The difference between B/A and b/a; should be as near zero as possible.*

If, on the other hand, we adopt the "ratio" interpretation, we have:

POSTULATE I. *The ratio between A/a and B/b (or the ratio between a/A and b/B; or the ratio between A/B and a/b; or the ratio between B/A and b/a; all of which have the same value) should be as near unity as possible.*

Since there is no way of choosing, mathematically, between Postulates Ia and Ib or between Postulates Ic and Id, and since these four demands lead to four different results, we shall reject all four of them and *adopt Postulate I as the proper interpretation of our Fundamental Principle.*

The case of two states is thus disposed of.

For the case of three or more states, one further principle is required, which we state as follows:

POSTULATE II. *In a satisfactory apportionment, there should be no pair of states which is capable of being "improved" by a transfer of representatives within that pair*—the word "improvement" being understood in the sense implied by the test already adopted for the case of two states, and the rare cases of "no choice" being decided in favor of the larger state.

From these two postulates the following theorem can then be deduced:

THEOREM I. *For any given values of A, B, C, ... and N, there will always be one and only one satisfactory apportionment in the sense defined by Postulates I and II.* No further principles are required.

A working rule for computing this "best" apportionment in any given case is found to be as follows:

Working Rule.—Multiply the population of each state by as many of the numbers

$$\text{Inf.}, 1/\sqrt{1 \times 2}, 1/\sqrt{2 \times 3}, 1/\sqrt{3 \times 4}, \ldots$$

as may be necessary, and record each result, together with the name of the state, on a small card. Arrange these cards according to the magnitude of the numbers recorded upon them, from the largest to the smallest, thus forming a *priority list* for the given states (the cards marked "Inf." being placed at the head of the list, arranged among themselves in order

of magnitude of the populations of the states). Finally, *assign the representatives, from the 1st to the Nth, to the several states in the order in which the names of the states occur in this priority list.* (It should be noted that this method satisfies automatically the constitutional requirement that every state shall have at least one representative.)

This method may be called the "*method of the geometric mean,*" since the "multipliers" are the reciprocals of the geometric means of consecutive integers.

The solution of the problem is thus complete.

Alternative Methods.—If we had adopted Postulate I*a* or I*b* we should have been led, in like manner, to two other methods which may be called the *method of the harmonic mean* (I*a*), and the *method of the arithmetic mean* (I*b*), since the "multipliers" in the working rules are as follows:

(I*a*)	Inf.,	$\frac{1+2}{2(1\times 2)}$,	$\frac{2+3}{2(2\times 3)}$,	$\frac{3+4}{2(3\times 4)}$,	...
(I*b*)	2,	2/3,	2/5,	2/7,	...

It can be shown that method I*a* favors the smaller states more than method I does, while method I*b* favors the larger states more than method I does. Since there is no mathematical reason for adopting either of the two Postulates I*a* and I*b* to the exclusion of the other, both should be rejected.

Postulates I*c* and I*d* also determine two distinct methods, which may be called *the two methods of similarity ratios.* It can be shown that I*c* favors the small states even more than I*a* does, while I*d* favors the large states even more than I*b* does, so that both should be rejected.

Each of these four methods violates three of the four conditions expressed in our Fundamental Principle, while the method of the geometric mean satisfies all these conditions simultaneously.

The following further methods are suggested by the Theory of Least Squares.

In a theoretically perfect apportionment, A/a would be equal to P/N, and a/A to N/P (where P is the total population). Hence, in place of Postulates I and II, we might consider the following:

POSTULATE III*a*. *The sum of the squares of the deviations of the A/a from their true values;* or

POSTULATE III*b*. *The sum of the squares of the deviations of the a/A from their true values; should be a minimum.*

It can be shown, however, that III*a* favors the smaller states even more than I*c* does, while III*b* favors the larger states even more than I*d* does. In other words, Postulates III*a* and III*b* violate, in opposite directions, all four of the conditions expressed in our Fundamental Principle. Since there is no mathematical reason for adopting either to the exclusion of the other, both should be rejected.

The same remark applies if in Postulates III*a* and III*b* the word "square" is replaced by "absolute value."[2]

Hence it is clear that if a *numerical measure of injustice* is desired, both the deviation of A/a and the deviation of a/A should be taken into account together. That is, any formula which reports A/a, say, as too large by a certain amount, should also report a/A as too small by the same amount. The formulas suggested by the simple application of the idea of least squares, as shown in the preceding paragraph, do not have this property. A combination of these formulas suggests, however, the following postulate, in which α denotes the theoretical value of a.

POSTULATE III. *In a satisfactory apportionment, the sum T of terms of the form Ae^2 where*

$$e = \frac{(a/A) - (\alpha/A)}{\sqrt{(\alpha/A)(a/A)}} = \frac{(A/a) - (A/\alpha)}{\sqrt{(A/\alpha)(A/a)}} = \frac{a - \alpha}{\sqrt{a\alpha}}$$

should be a minimum.[3]

For purposes of computation, this total error, T, may be replaced by an average error, $E = \sqrt{S/N}$, where S is the sum of terms of the form $\alpha e^2 = (a - \alpha)^2/a$.

The method determined by Postulate III is precisely the same as the method of equal proportions based on Postulates I and II.

[1] This article contains the substance of two papers presented to the American Mathematical Society, December 28, 1920, and February 26, 1921. Further details, with proofs and examples, will be published either in the Transactions of the American Mathematical Society, or in the Quarterly Publication of the American Statistical Association, or in the American Mathematical Monthly.

For the history of the subject see W. F. Willcox, "The Apportionment of Representatives" (presidential address at the annual meeting of the American Economic Association, December 1915), published in the American Economic Review, Vol. 6, No. 1, Supplement, pp. 1–16, March, 1916. See also 62d Congress, 1st Session, House of Representatives, Report No. 12, pp. 1–108, April 25, 1911, and John H. Humphreys, "Proportional Representation," London, 1911.

The most important of the methods hitherto known are four:

The *Vinton method of 1850*, long in use in Congress, is known to lead to an "Alabama paradox;" that is, an increase in the total size of the House may cause a decrease in the representation of some state.

The *Hill method of alternate ratios*, proposed by Dr. J. A. Hill in 1910 but not adopted, comes very near to satisfying the postulates of the present paper, and uses for the first time (though only partially) the idea of the geometric mean. The method is incomplete however, since it can be shown to lead to an Alabama paradox.

The *Willcox method of major fractions*, devised by Professor W. F. Willcox in 1900–1910, and now in use in Congress, employs, in effect, a working rule like ours with multipliers: Inf., 2/3, 2/5, 2/7, ...; it is essentially the same as the method of the arithmetic mean, and therefore favors the larger states unduly, just as the hitherto unsuspected but equally justifiable method of the harmonic mean favors the smaller states unduly. (It may be noted that the name "major fractions" is somewhat misleading, since the Willcox major fraction is not a major fraction of the true quota, but

a major fraction of an artificial quota, scaled up or down from the true quota to meet the requirements of the computation.)

The *d'Hondt method*, originated in Belgium and now widely used in European elections, employs "multipliers" 1, 1/2, 1/3, 1/4... and can be shown to favor the larger states to the extent of violating all four of the conditions expressed in our Fundamental Principle.

[2] If one should try to minimize the *sum of the squares* (or the sum of the absolute values) *of the deviations of the a's themselves* from their true values (with or without "weighting" by the population of the state), the resulting methods would all lead to an Alabama paradox. The same is true of the weighted sum of the absolute values of the deviations of a/A (or of A/a). The same is also true of the absolute values of the logarithms of the ratios between the a'S and their true values.

[3] This Postulate III was added on April 23, after Professor F. W. Owens had shown (at the meeting of the American Mathematical Society on February 26) that the method of minimizing the sum of terms like $A[(a/A) - (\alpha/A)]^2$ leads to th same result as the Willcox method of major fractions. It may be noted that the method of minimizing the sum of terms like $a[(A/a) - (A/\alpha)]^2$ leads, not as one might expect, to the method of the harmonic mean, but to the method of the geometric mean.

CURRENT MAPS OF THE LOCATION OF THE MUTANT GENES OF DROSOPHILA MELANOGASTER[1]

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The maps that have been published[2] showing the distribution of the mutant genes of D. melanogaster can now be much improved because of the discovery of new mutants and the accumulation of crossover data. Figure 1 gives in simplified form the maps that are in use in our laboratory.

The distances on the maps are based on the total amount of crossing over between the loci, one unit of distance representing one per cent of crossing over. The map-distances are the same as the observed crossover values or "percentages of exchange" whenever the two loci considered are so close together that no, or only a negligible amount of, double crossing over occurs between them. In the first (X-) chromosome this practical equivalence of map-distance and exchange-value holds for loci not farther apart than about 15 units. In the middle of the second chromosome and of the third chromosome the equivalence holds for only about 10 units. In the end-regions of the second and third chromosomes it holds up to nearly 20 units. For distances somewhat greater than these the map-distances exceed the observed percentages of exchange by an amount equal to twice the percentage of double crossing over between the given loci. For still greater distances the difference includes also three times the percentage of triples. The number of quadruple crossovers is negligible except perhaps when the whole length of the second chromosome is to

be considered. For each chromosome and for each region within a chromosome the amount of this multiple crossing over is characteristic, and may not be the same in amount for different sections of equal map-distance. Because of this variation, the accurate expression of the relation between map-distance and exchange-values for the more distant loci, requires a table of conversion corrections for each pair of loci considered. Such tables will be published with the more detailed maps of the chromosomes. In general, the correction is relatively slight with distances that are under 20–40, but for longer intervals the correction increases at an accelerated rate. We have not met with percentages of exchange that exceed 50.0, though two of the maps are about a hundred units long.

The map of the third chromosome is the most accurate, since the calculation of the distances between the principal loci is made on the basis of all data up to 1920, and an improved method of weighting and interrelating the data has been followed. Relatively little change in these primary distances—the "triangulation" of the map—is expected with the further accumulation of data. There is still some uncertainty with regard to the region to the left of spineless, for the different sets of data upon that region may not be comparable because of the possible presence of crossover variations. The data used in the calculations for the primary distances in the first and second chromosomes are homogeneous, and although not including the last four years' work, are still fairly ample in amount. Changes are to be expected when these two maps are recalculated with the improved method and complete data.

The most useful mutants are those that are separable from the wild-type with completeness and ease, that are not inferior to the wild-type in viability and productivity, and that do not interfere with the use in the same experiment of any large class of the other mutant characters. Dominants are more valuable than recessives. Mutants accurately located in the chromosome are more valuable than those whose positions are less well established, though if the other desirable features mentioned above are present in a given new mutant the position will be found rather quickly. A very large factor in the value of the mutant is its position in the chromosome. The end positions are most valuable. Throughout the remainder of the chromosome the most favorable positions are those evenly spaced and just close enough together so that no double crossing over occurs between them. If the interval is too small there is trouble in getting double recessives, and the crossover classes are so small that large totals are required to make differences significant.

The mutants that fulfill all the above requirements most nearly are followed by an asterisk (*) in the maps. Their loci are the primary bases with relation to which the other mutants are located. There are several cases in which two or more excellent mutants affecting quite different

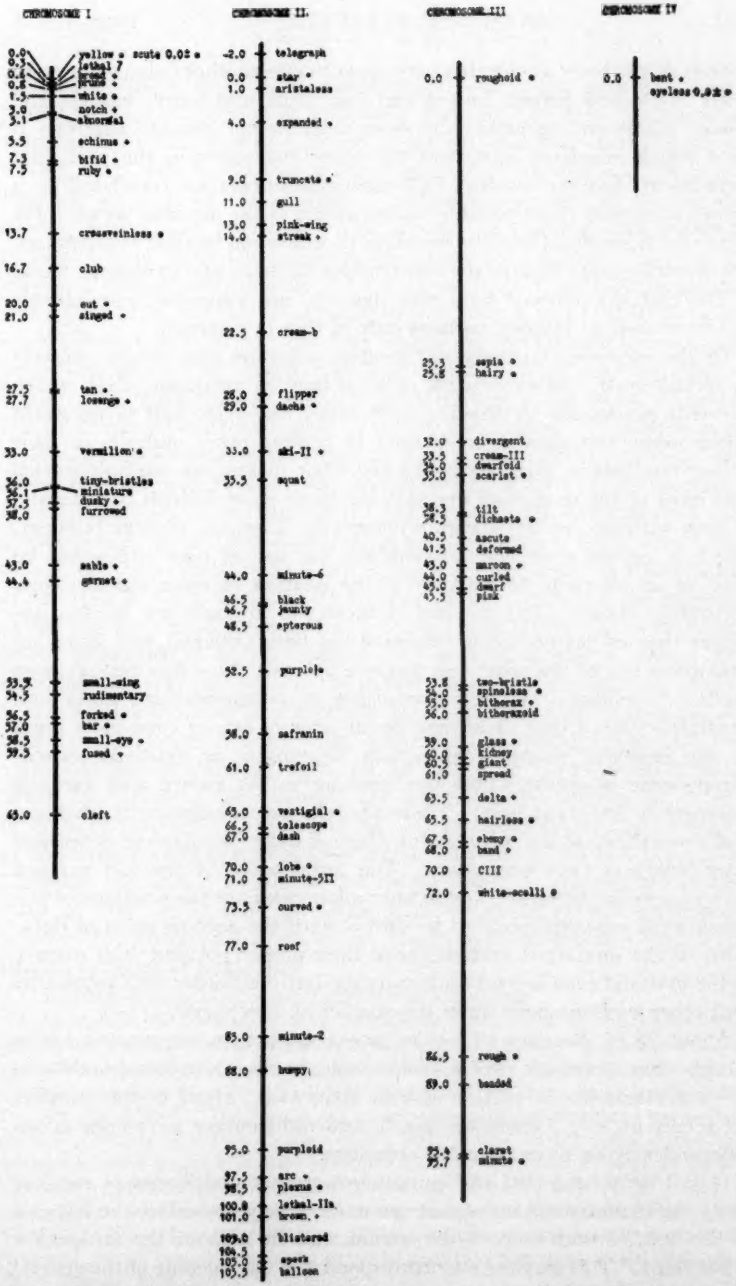


FIG. 1. Maps showing the distribution of mutant loci in the four chromosomes of *Drosophila melanogaster* (Oct. 15, 1920).

[A † sign should be added to miniature (I,36.1) and gull (II,11.0), and a * sign to black (II,46.5), speck (II,105.0), pink (III,45.5), and claret (III,95.4).]

regions of the body are located very close to one another (e. g., yellow and scute, sable and garnet, forked and bar, sepia and hairy, hairless and ebony, claret and minute). Of these neighboring mutants that one is used which interferes least with the other characters in the particular experiment, and the results of all such experiments are correlated by a simple correction made possible by knowledge of the distance between the loci. Such groups of loci, the members of which can be used as alternates, are treated as one base in the construction of maps and in linkage work.

The mutants followed by a plus sign (+) are somewhat less valuable in one or another respect, perhaps only in that of position.

Of the remaining mutants, not marked, some are exceedingly valuable in special work. As an example of these may be mentioned CIII, which prevents practically all crossing over within the right half of the third chromosome and thus is much used in holding other mutants in their proper relations in balanced stocks and while making up multiple stocks. But most of the unmarked mutants are those more difficult to separate, of poor viability, or less accurately located. Thus the mutant bithorax, which is located very close to spineless, has second rank and would be used as an alternate to spineless if the relation between the loci were accurately known. But the loci of these two mutants are so close together that as yet no double recessive has been obtained, and hence no back-cross test of the order and distance between these two loci has been made. Dominants, and recessives close to dominants, are easily and accurately placed, but there may be an error of five or even more units in the indicated positions of certain (unmarked or omitted) second-chromosome recessives whose positions are as yet known only through reference to a distant base. These are relatively unimportant mutants, and since they are not very useful, there is little incentive to determine their locations more accurately. The serial order of the loci marked with an asterisk is certain, and in only a few cases are the positions of the second-rank mutants likely to be shifted with the accumulation of data. Most of the unmarked mutants have their proper position with respect to the first and even second rank mutants, but their order with respect to each other is often uncertain in the absence of direct tests.

About 25 of the mapped loci represent more than one mutant allelomorph; thus, there are twelve allelomorphs at the white locus, and three to six allelomorphs at each of several other loci. Many of the mutants are recurrent; e. g., vermilion, notch, and rudimentary have each arisen independently on more than ten occasions.

It will be noticed that the mutant loci are not distributed at random along the chromosome maps, but are more closely spaced at the left end of the first, at both ends of the second, and throughout the mid-region of the third. This massing may correspond to a like massing of the genetic

materials. When orthopteran chromosomes are in the extended phase preceding their condensation as tetrads they may be seen to consist of granules of unequal size distributed at unequal distances along the linin thread, both granule size and spacing being characteristic and relatively constant for a given chromosome. Another explanation of the uneven spacing seems more probable: The amount of crossing over between given loci, e. g., black and purple in the second chromosome, is known to vary in a definite manner with the age of the mother, with temperature, and with genetic modifiers of crossing over. Since a unit of map-distance is thus known to represent different lengths of chromosome thread, the regions in which loci appear relatively closely spaced may be regions in which one unit of map distance represents a longer section of chromosome than in those regions that appear relatively unoccupied. It may be supposed that each region of chromosome has its characteristic relation between unit of map-distance and corresponding length of chromosome. It is probably significant that in the second and third chromosomes the distribution of mutant loci is approximately symmetrical about their midpoints, for these chromosomes are V-shaped with median attachment of the spindle fibre, while the first chromosome, which has the massing at one end, has terminal attachment. The variations, from region to region, in the proportion of double crossing over, referred to above, show a distribution that is symmetrical in the second and third chromosomes. From this and other evidence it is probable that there is a definite relation between the region of chromosome, considered morphologically, and the variations in the ratio of map-unit to length of chromosome thread. When the different chromosomes are compared the average ratios are found to be practically the same for all. Thus, the lengths of chromosomes as measured directly in metaphase plates are in the ratio of 100 : 159 : 159: 12, while the known lengths of the chromosomes as mapped are in the ratio of 100 : 165 : 155 : 2.[3] This is a surprisingly close agreement between the observed lengths of the condensed chromosomes and their lengths in terms of map-units.

[1] Contribution from the Carnegie Institution of Washington.

[3] The maps in the frontispiece of the Mechanism of Mendelian Heredity were made in 1914, and are both incomplete and out of proportion. A new frontispiece, practically the same as the figure accompanying this paper, has been supplied for a revision of the Mechanism and for a French edition. A similar figure is to appear in a book by Dr. Sharp of Cornell. The best of the earlier maps of the first chromosome is that given in *Carnegie Institution Publication* No. **237**, p. 22. Partial maps of the first chromosome that are more modern have been given in connection with special papers (*Genetics*, **1**, 1916 (8); *Amer. Nat.*, **1916** (421); *J. Gen. Physiol.*, **1919** (646). A map of the second chromosome, complete to 1916, is given in *Carnegie Institution Publication* No. **278**, p. 303. No satisfactory map of the third chromosome has previously been published, though partial maps based on special work have been given (*Amer. Nat.*, **1916** (217); *Genetics*, **1919** (208).

[1] Since figure 1 was drawn (Oct. 15, 1920) a mutant (M-23) has been located at 101 in chromosome III, and there has been found to be about 1% of crossing over between bent and eyeless in the fourth chromosome. (Scarlet is at 41.5, Chromosome III.)